

The Law of Sines

Solving a Triangle: Finding all three side lengths and all three interior angle measures of a triangle.

Scenarios when Solving Triangles

The Law of Cosines

SSS SAS

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = c^2 + a^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

The Law of Sines

AAS

ASA

SSA

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

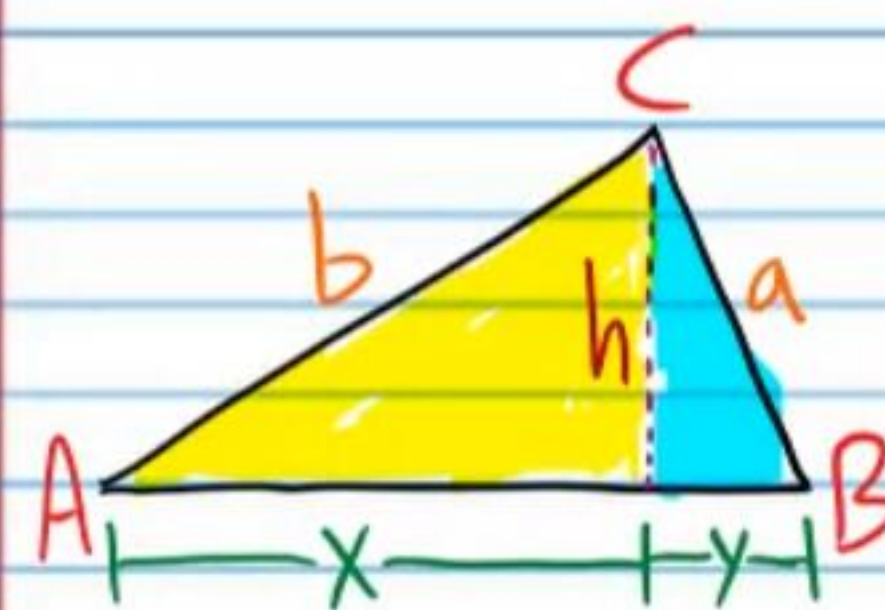
Ambiguity

i) No triangle

ii) One triangle

iii) Two triangles

A Short Explanation of the Law of Sines



$$\sin A = \frac{h}{b}$$

$$\sin B = \frac{h}{a}$$

$$b \sin A = h, \quad a \sin B = h$$

$$h = h$$

$$b \sin A = a \sin B$$

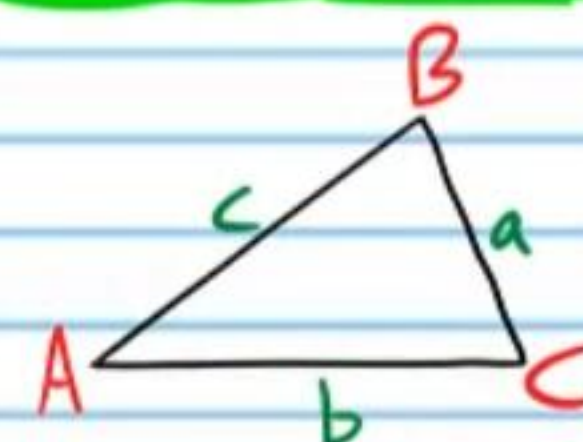
$$\frac{b \sin A}{ab} = \frac{a \sin B}{ab}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

The Law of Sines

If a , b , and c are the side lengths of a triangle, and A , B , and C are the interior angle measures, then the equation to the right is true.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



The Angle-Angle-Side Scenario (AAS)

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

- ① I) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$20^\circ + 18^\circ + C = 180^\circ$$

$$\boxed{C = 142^\circ}$$

- II) Find a .

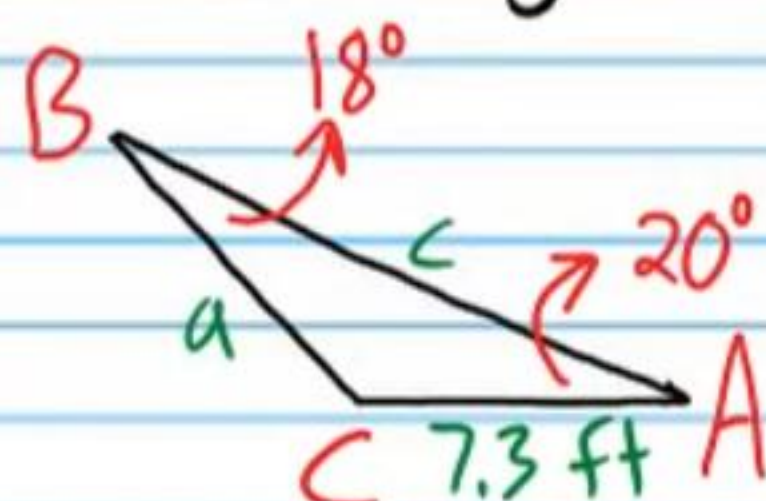
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 20^\circ}{a} = \frac{\sin 18^\circ}{7.3}$$

$$\frac{(7.3)\sin 20^\circ}{(7.3)a} = \frac{\sin 18^\circ}{7.3} (a)$$

$$\frac{7.3\sin 20^\circ}{7.3a} = \frac{a\sin 18^\circ}{7.3a}$$

$$7.3\sin 20^\circ = a\sin 18^\circ$$



$$\frac{7.3\sin 20^\circ}{\sin 18^\circ} = \frac{a\sin 18^\circ}{\sin 18^\circ}$$

$$\frac{7.3\sin 20^\circ}{\sin 18^\circ} = a$$

$$\boxed{8.08 \text{ ft} \approx a}$$

- III) Find c .

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

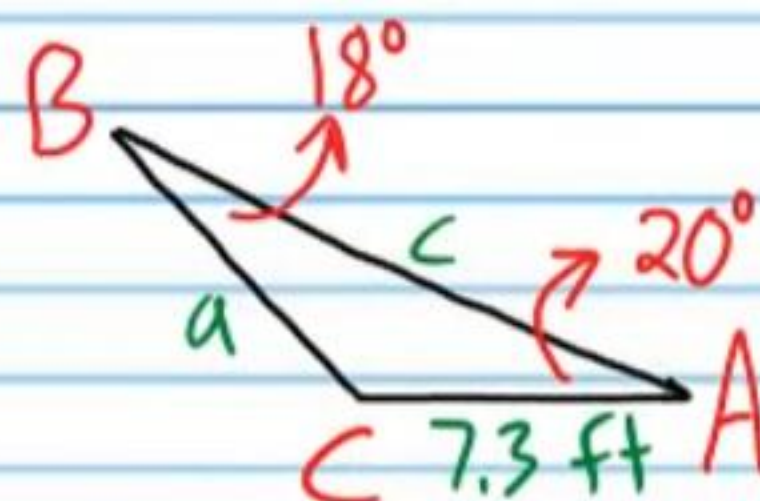
$$\frac{\sin 18^\circ}{7.3} = \frac{\sin 142^\circ}{c}$$

$$c\sin 18^\circ = 7.3\sin 142^\circ$$

$$\frac{c\sin 18^\circ}{\sin 18^\circ} = \frac{7.3\sin 142^\circ}{\sin 18^\circ}$$

$$c = \frac{7.3\sin 142^\circ}{\sin 18^\circ}$$

$$\boxed{c \approx 14.54 \text{ ft}}$$

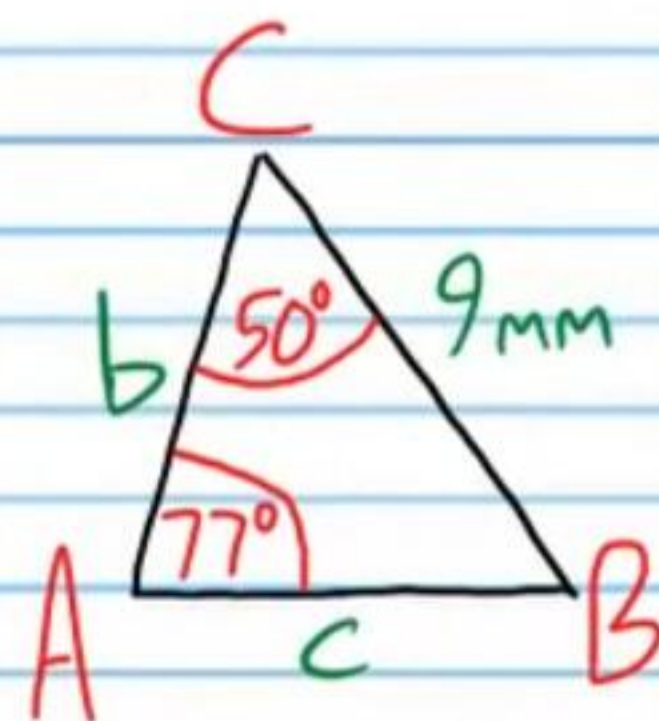


② I) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$77^\circ + B + 50^\circ = 180^\circ$$

$$\boxed{B = 53^\circ}$$



II) Find c .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 77^\circ}{9} = \frac{\sin 50^\circ}{c}$$

$$(9c) \frac{\sin 77^\circ}{9} = \frac{\sin 50^\circ}{c} (9c)$$

$$c \sin 77^\circ = 9 \sin 50^\circ$$

$$\frac{c \sin 77^\circ}{\sin 77^\circ} = \frac{9 \sin 50^\circ}{\sin 77^\circ}$$

$$c = \frac{9 \sin 50^\circ}{\sin 77^\circ}$$

$$\boxed{c \approx 7.08 \text{ mm}}$$

III) Find b .

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 77^\circ}{9} = \frac{\sin 53^\circ}{b}$$

$$\frac{9}{\sin 77^\circ} = \frac{b}{\sin 53^\circ}$$

$$\sin 53^\circ \frac{9}{\sin 77^\circ} = \frac{b}{\sin 53^\circ} \sin 53^\circ$$

$$\frac{9 \sin 53^\circ}{\sin 77^\circ} = b$$

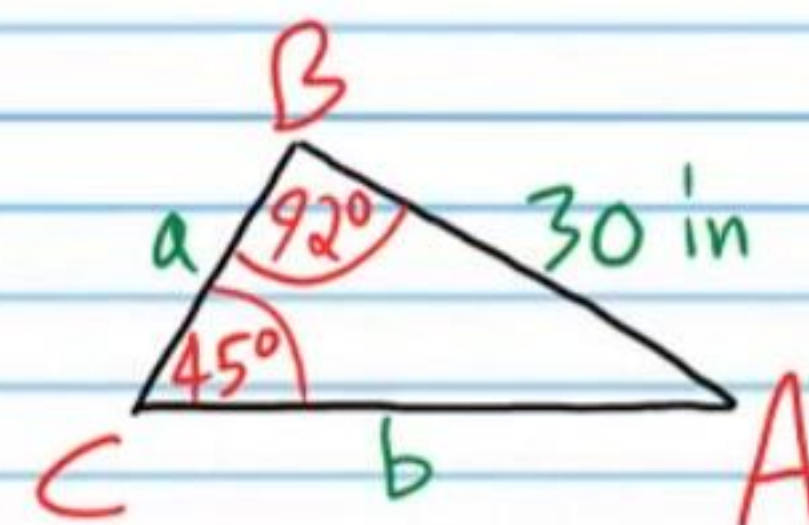
$$\boxed{7.38 \text{ mm} \approx b}$$

③ I) Find $m\angle A$.

$$A + B + C = 180^\circ$$

$$A + 92^\circ + 45^\circ = 180^\circ$$

$$\boxed{A = 43^\circ}$$



II) Find b .

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 92^\circ}{b} = \frac{\sin 45^\circ}{30}$$

$$b = \frac{30 \sin 92^\circ}{\sin 45^\circ}$$

$$\boxed{b \approx 42.4 \text{ in}}$$

III) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 43^\circ}{a} = \frac{\sin 45^\circ}{30}$$

$$a = \frac{30 \sin 43^\circ}{\sin 45^\circ}$$

$$\boxed{a \approx 28.93 \text{ in}}$$

④ I) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$10^\circ + 22^\circ + C = 180^\circ$$

$$\boxed{C = 148^\circ}$$

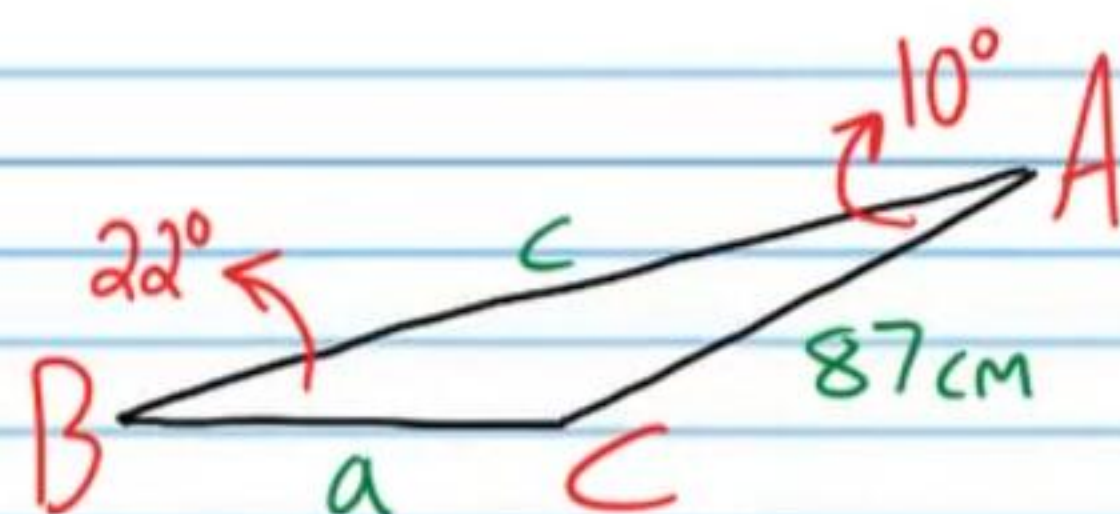
II) Find a .

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 10^\circ}{a} = \frac{\sin 22^\circ}{87}$$

$$a = \frac{87 \sin 10^\circ}{\sin 22^\circ}$$

$$\boxed{a \approx 40.33 \text{ cm}}$$



III) Find c .

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 148^\circ}{c} = \frac{\sin 22^\circ}{87}$$

$$c = \frac{87 \sin 148^\circ}{\sin 22^\circ}$$

$$\boxed{c \approx 123.07 \text{ cm}}$$

The Angle-Side-Angle Scenario (ASA)

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

• ⑤ I) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$95^\circ + B + 31^\circ = 180^\circ$$

$$\boxed{B = 54^\circ}$$

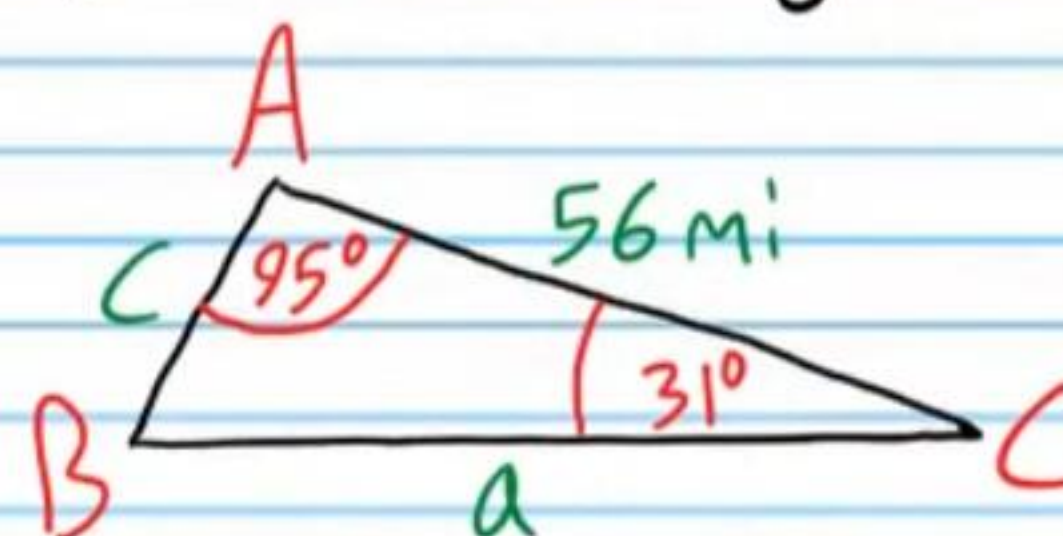
II) Find c .

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 31^\circ}{c} = \frac{\sin 54^\circ}{56}$$

$$c = \frac{56 \sin 31^\circ}{\sin 54^\circ}$$

$$\boxed{c \approx 35.65 \text{ mi}}$$



III) Find a .

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 95^\circ}{a} = \frac{\sin 54^\circ}{56}$$

$$a = \frac{56 \sin 95^\circ}{\sin 54^\circ}$$

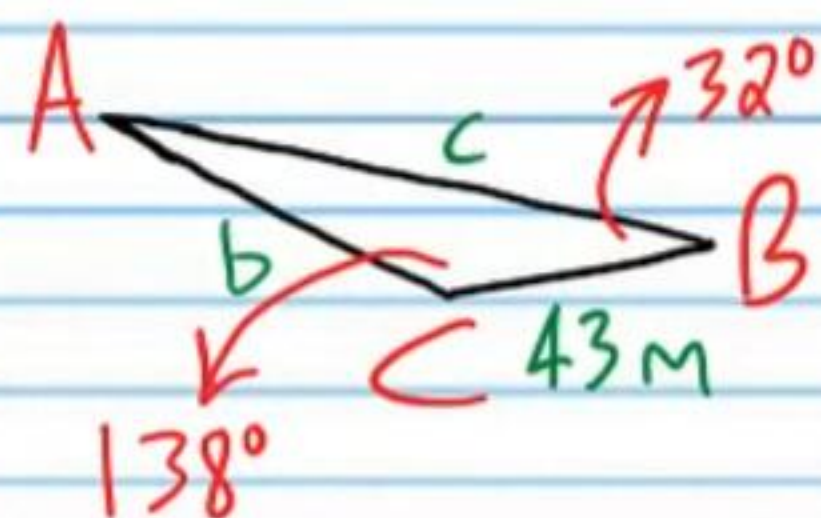
$$\boxed{a \approx 68.96 \text{ mi}}$$

- ⑥ I) Find $m\angle A$.

$$A+B+C=180^\circ$$

$$A+32^\circ+138^\circ=180^\circ$$

$$\boxed{A=10^\circ}$$



- II) Find b .

$$\frac{\sin B}{b} = \frac{\sin A}{a}$$

$$\frac{\sin 32^\circ}{b} = \frac{\sin 10^\circ}{43}$$

$$b = \frac{43 \sin 32^\circ}{\sin 10^\circ}$$

$$\boxed{b \approx 131.22 \text{ m}}$$

- III) Find c .

$$\frac{\sin C}{c} = \frac{\sin A}{a}$$

$$\frac{\sin 138^\circ}{c} = \frac{\sin 10^\circ}{43}$$

$$c = \frac{43 \sin 138^\circ}{\sin 10^\circ}$$

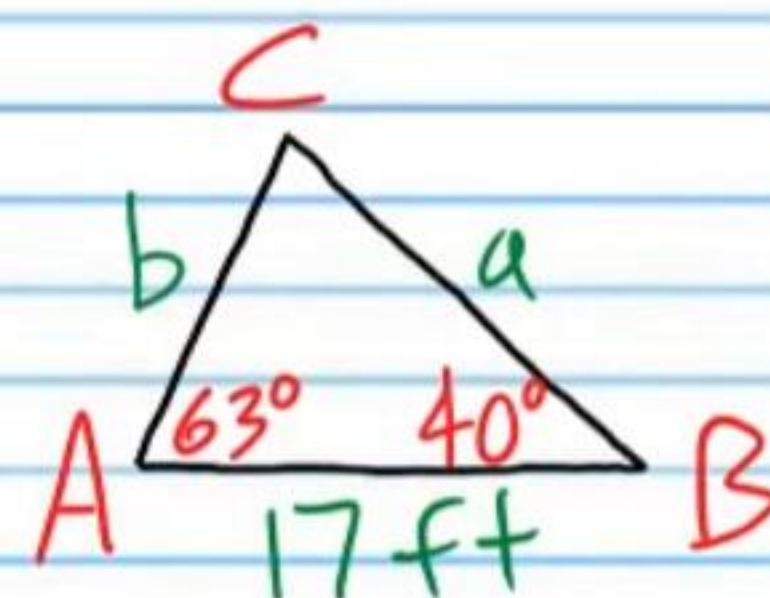
$$\boxed{c \approx 165.69 \text{ m}}$$

- ⑦ I) Find $m\angle C$.

$$A+B+C=180^\circ$$

$$63^\circ+40^\circ+C=180^\circ$$

$$\boxed{C=77^\circ}$$



- II) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 63^\circ}{a} = \frac{\sin 77^\circ}{17}$$

$$a = \frac{17 \sin 63^\circ}{\sin 77^\circ}$$

$$\boxed{a \approx 15.55 \text{ ft}}$$

- III) Find b .

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 40^\circ}{b} = \frac{\sin 77^\circ}{17}$$

$$b = \frac{17 \sin 40^\circ}{\sin 77^\circ}$$

$$\boxed{b \approx 11.21 \text{ ft}}$$